Fall 2023 / EC571

Instructor: Ajay Joshi

Mid-Term Exam 1

Date: October 10, 2023

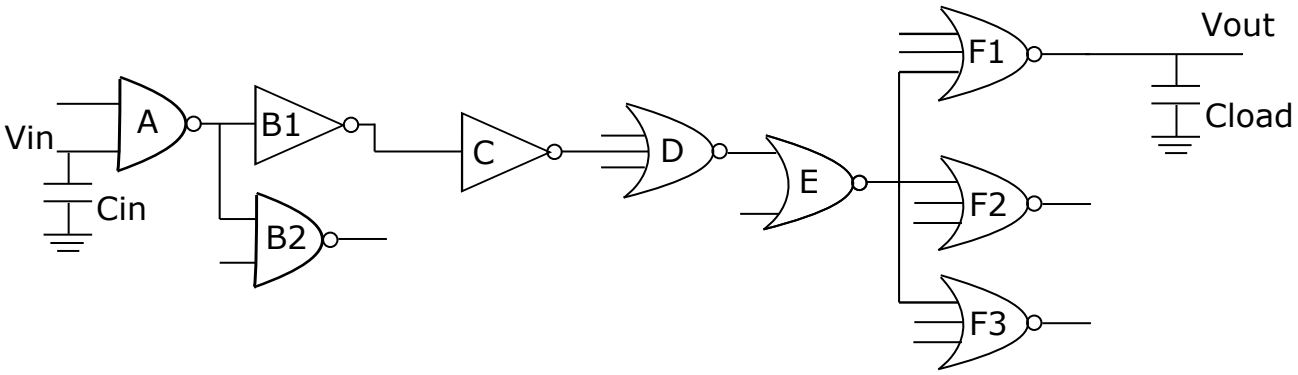
Name : _____

BU ID : _____

Question	Maximum points	Points scored
1	20	
2	20	
3	20	
4	20	
5	20	
6	20	
Bonus	5	
Total	100	

1. The duration of the exam is 1 hour 45 minutes.
2. The maximum grade is 100 points.
3. Please write your name on all pages.
4. Attempt any 5 of the 6 questions. Please indicate the question number that should not be graded.
5. If you need to use extra sheets of paper, write your name and BU ID on it, and clearly indicate the question number.
6. Material sharing is not allowed.
7. Calculators are allowed.
8. The exam is closed book close notes, but you are allowed to have two cheat sheets.
9. Boston University Academic Code of Conduct applies.
10. BU ENG is collaborating with several online services, including Chegg, to monitor activity during exams and obtain account information and IP addresses connected to uploaded exam materials and their access. The penalties for cheating are extremely severe including expulsion from BU.

Question 1



For the circuit shown in the above figure, using the logical effort technique, calculate the width of the transistors for each gate in the path such that the total delay of the path is minimized. Note that B1 and B2 are not identical. You can assume that gates F1, F2, and F3 are identical. Assume C_{in} = 5 fF, C_G = 1 fF/μm, C_{load} = 100 fF and $2R_{sq(NMOS)} = R_{sq(PMOS)}$. (20 points)

Question 2

1. Implement the following function $F = \overline{(P \cdot Q + R) \cdot (S + T)}$ with the static complementary CMOS logic. Assume $R_{sq,PMOS} = 2 * R_{sq,NMOS}$, and size your transistors so that the worst-case pull-up resistance is equal to the worst-case pull-down resistance. (10 points)

2. Using the switch model for the transistors, draw the RC network you would use to calculate the delay of the gate calculating the above function F when $P = V_{DD}, Q = V_{DD}, R = 0V, S = 0V$ and T changes from $0V$ to V_{DD}. You can assume $C_G = 1fF/\mu$, $C_D = 1.5fF/\mu$, $R_{sq(NMOS)} = 10K\Omega/sq$, and $R_{sq(PMOS)} = 20K\Omega/sq$. Assume $C_L = 50fF$. Include all parasitic capacitances. (7 points)

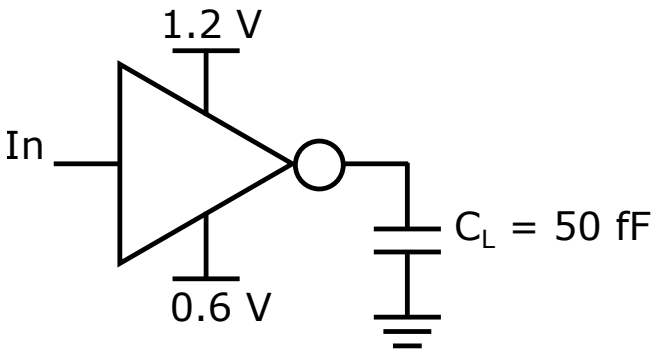
3. Determine the delay of the implementation above for the same inputs. Provide your answer for the delay in *ps*. (3 points)

Question 3

1. What are the three different regions of operations of a typical NMOS transistor? (5 points)

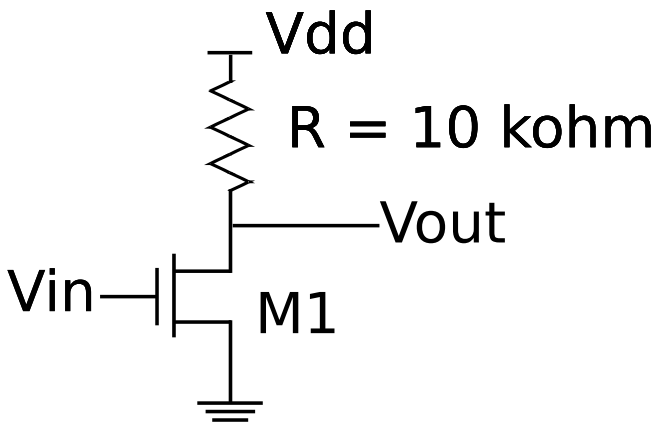
2. Write down the current equations for the three regions of operation for a long-channel NMOS transistor. (5 points)

3. For the inverter shown below how much energy is consumed by the inverter when the input switches from 1.2 V to 0 V and then back up to 1.2 V. You can ignore all the capacitors associated with the transistors inside the inverter. (5 points)



4. Explain the concept of velocity saturation in your own words. (5 points)

Question 4



Assume V_{dd} = 2.5 V, $V_{TH,n}$ = 0.5 V and no short-channel effects.

1. For V_{in} = 0 V, what is the value of V_{out}? (2 points)

2. For V_{in} = 2.5 V and V_{out} = 2.5 V, what is the region of operation for M1? (3 points)

3. Write the equation for calculating V_{out} in that region (3 points)

4. With V_{in} fixed at 2.5 V, V_{out} will start reducing. At what output voltage does the region of operation change for M1? (3 points)

5. Write the equation for calculating V_{out} for this new region of operation (3 points)

6. As V_{out} further decreases, what happens when $V_{out} << (V_{in} - V_{TH,n})$ (3 points)

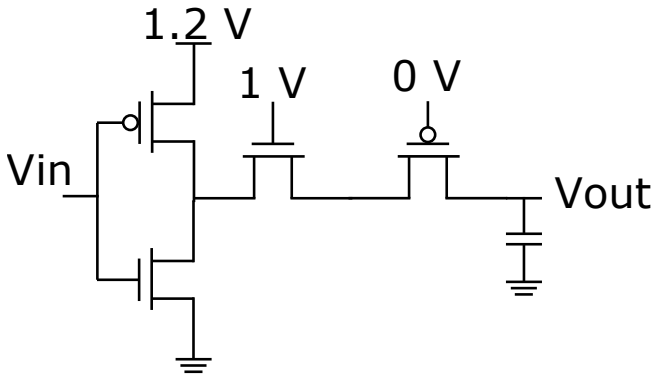
7. Write the equation for calculating V_{out} for this new region of operation (3 points)

Question 5

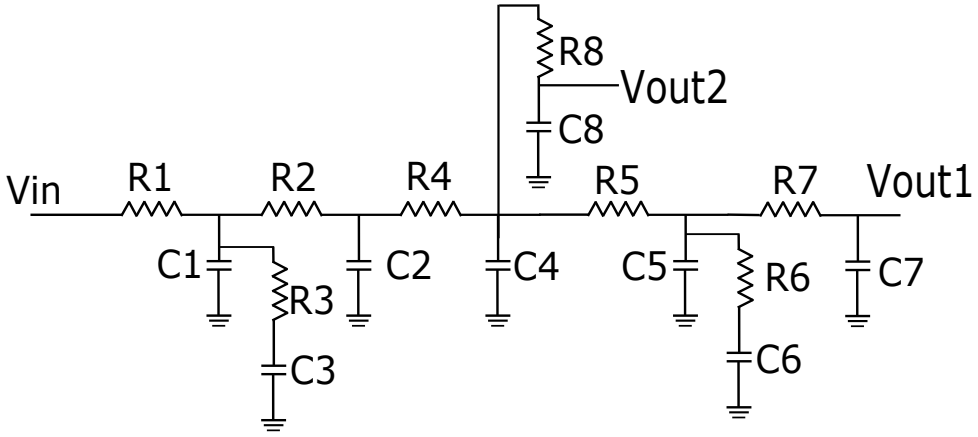
Design the static complementary CMOS implementation of a 1-bit half-adder circuit operating at 2 GHz with both sum and carry outputs driving a load of 100 fF. As part of the design, you need to determine the width of all transistors. You can use the following transistor parameters for the design - L = 0.25 micron $R_{sq}(NMOS)$ = 13 Kohm/sq, $R_{sq}(PMOS)$ = 26 Kohm/sq, C_g = 1.5 fF/micron, C_d = 2 fF/micron. You can assume that you have both complemented and non-complemented inputs and that input signals have very small rise and fall times. *Hint: A 1-bit half adder has two 1-bit inputs – A and B and it generates two 1-bit outputs – SUM and CARRY.* (20 points)

Question 6

1. Using the simple switch model of the transistors with $R_{sq(NMOS)} = 10K\Omega/sq$,, $R_{sq(PMOS)} = 20K\Omega/sq$, $V_{TH,N} = 0.3V$ and $V_{TH,P} = -0.3V$, draw the VTC for the circuit shown below. Assume all transistors have a width of 1 μm If the circuit is digital, provide the values of V_{OH}, V_{OL}, V_{IH}, and V_{IL}; if the circuit is not digital, explain why it isn't digital. (15 points)



2. Write the RC delay equation for Vout1 and Vout2 using the Elmore delay approach for the circuit shown below. (5 points)



Bonus question

Prof. Joshi is thinking of visiting someplace warm during the winter vacation. Suggest 5 places he can visit. (5 points)